

Magnetic Support and Stable Field Geometry Are Mutually Exclusive in Super-Chandrasekhar White Dwarfs

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ABSTRACT

Equilibrium models of magnetized white dwarfs above the Chandrasekhar limit support the extra mass with an ordered interior field carrying a substantial fraction of the gravitational binding energy. We ask whether such a star can carry that field at all, and whether the field can be given the mixed poloidal–toroidal geometry that is the only one known to be dynamically stable. Using self-consistent-field equilibria with a Grad–Shafranov poloidal component and a confined toroidal component, we survey 1.8×10^3 configurations spanning central density, field strength, field localization, and rotation. We find a ceiling on the confined toroidal field that the star can hold: $E_{\text{tor}}/|W| \leq 0.0101$, essentially independent of central density over a decade and converged to 0.01% between 129^2 and 257^2 meshes. The equilibrium literature requires $E_{\text{tor}}/|W| \simeq 0.203$, twenty times more. A field the star can hold is worth $+0.017 M_{\odot}$, about one per cent, and the mass frontier converges to M_{Ch} from below without crossing it, terminating at neutronization. Adding differential rotation removes the mass restriction but not the field ceiling: configurations above the Chandrasekhar mass with comparable poloidal and toroidal energies exist, reaching $2.13 M_{\odot}$ at the edge of the comparable band and $1.65 M_{\odot}$ well inside it, but their mass is rotational and their field still carries only $\sim 1\%$ of the virial. Three-dimensional ideal-MHD evolution of the configuration the literature specifies confirms that it is destroyed by an $m = 1$ instability within a few Alfvén times. We further show that the stable mixed configuration cannot be evolved in a domain-filling code at accessible cost: comparable poloidal energy implies an exterior dipole of $\sim 10^{11}$ – 10^{12} G, which no affordable ambient medium can confine. We argue that super-Chandrasekhar white dwarfs, if they exist, are rotationally supported objects in which the magnetic field is a passenger rather than the support.

Keywords: White dwarfs (1799) — Stellar magnetic fields (1610) — Magnetohydrodynamics (1964) — Stellar rotation (1629) — Type Ia supernovae (1728)

1. INTRODUCTION

The possibility that a white dwarf can exceed the Chandrasekhar mass if it is strongly magnetized has been developed over the last decade into a programme of equilibrium models (e.g. Subramanian & Mukhopadhyay 2015; Gupta et al. 2020; Deb et al. 2022). The construction is consistent within its own assumptions: a sufficiently strong ordered interior field contributes pressure and tension that support mass beyond what electron degeneracy alone can hold, and sequences of such equilibria reach 2–2.3 $M_{\odot}$ for interior fields of a few times 10^{13} G.

Two things are absent from that programme, and they are connected.

The first is dynamics. The models are stationary equilibria; none has been evolved. This matters because the field geometries the sequences use — either

purely toroidal or purely poloidal — are precisely the two that linear theory says are unstable. Braithwaite & Spruit (2006) showed that a stable stellar field requires poloidal and toroidal components of *comparable* energy, the twisted-torus configuration anticipated by Prendergast (1956); both pure branches are unstable, the toroidal one to the $m = 1$ Tayler kink. The twisted torus does not appear anywhere in the super-Chandrasekhar white dwarf literature.

The second is the question of whether the star can hold the field it is given. A field that supports mass necessarily carries a large fraction of the binding energy, and therefore a magnetic pressure comparable to the gas pressure somewhere in the star. Where the plasma $\beta = P_{\text{gas}}/P_{\text{mag}}$ falls below unity, the field dominates the pressure that the non-magnetic equation of state

used to build the equilibrium was supplying, and the construction is inconsistent with itself.

This paper addresses both. Section 2 describes the equilibrium construction and the criteria we impose. Section 3 establishes a ceiling on the confined toroidal field a white dwarf can hold and shows that the mass frontier it permits saturates at M_{Ch} . Section 4 adds differential rotation and shows that super-Chandrasekhar twisted-torus configurations do exist, but that their mass is rotational. Section 5 presents three-dimensional evolution of the literature’s configuration and of the stable alternative. Section 6 draws out what this implies for the programme, and Section 7 states the limitations, several of which are severe and are placed there deliberately rather than at the end.

2. METHODS

2.1. Equilibrium construction

Equilibria are computed with a self-consistent-field scheme in the manner of Hachisu (1986), on a spherical (r, θ) grid without assumed equatorial symmetry. The equation of state is the zero-temperature degenerate electron gas of Chandrasekhar (1935) with $\mu_e = 2$, which is also the equation of state used in the three-dimensional evolution, so that the initial condition is in genuine equilibrium under the same physics that evolves it.

The toroidal field is self-consistent, entering the Bernoulli integral through its own term, with $B_\phi \propto \rho\varpi$. The poloidal field follows from the Grad-Shafranov equation

$$\Delta^* u = -4\pi\varpi^2 \rho \mu(u), \quad (1)$$

where the arbitrary function μ controls how the current is distributed. The choice $\mu = \text{const}$, used by essentially all earlier work, produces a vacuum dipole exterior. Following Fujisawa et al. (2012) we also use the negative-power form $\mu(u) \propto (u + \epsilon)^m$, which concentrates current toward the axis and enlarges the closed-field-line region; $m = 0$ recovers the constant case. Because μ depends on u the equation is nonlinear and is solved by under-relaxed iteration.

A twisted torus requires the toroidal field to vanish on every field line that reaches the vacuum, so the toroidal component is confined to the closed-line region, with the profile $B_\phi \propto w^\zeta/\varpi$ where w is the normalized flux inside that region. This confinement is the physical content of the twisted-torus requirement and, as we show below, it is also what limits the field the star can hold.

Rotation follows the j -constant law $\Omega(\varpi) = \Omega_c A^2/(A^2 + \varpi^2)$ with A set to the equatorial radius of the non-rotating reference star at the same central

density, and Ω_c quoted as a fraction of that reference’s Keplerian frequency.

2.2. The four criteria

A configuration is counted as viable only if it satisfies all of:

1. $M > M_{\text{Ch}} = 1.44 M_\odot$ for $\mu_e = 2$: it is super-Chandrasekhar at all.
2. $\beta_{\text{min}} \geq 1$, where β_{min} is the minimum of $P_{\text{gas}}/P_{\text{mag}}$ over the region $\rho > 10^6 \text{ g cm}^{-3}$: the star can hold its own field. We stress that this is a consistency condition on the construction, not a proof of dynamical stability.
3. $0.1 < E_{\text{tor}}/E_{\text{pol}} < 10$: the poloidal and toroidal energies are comparable, which is the Braithwaite & Spruit (2006) stable branch. Both pure branches are excluded by this criterion, as they should be.
4. $T/|W| < 0.14$ and no mass shedding: the rotation is one the star can keep. The threshold is the secular bar mode; the dynamical one is 0.27. A star held up by rotation it cannot retain is not a result.

Criterion (iv) is inactive for the non-rotating survey of Section 3 and becomes the binding constraint in Section 4.

2.3. Three-dimensional evolution

Evolution uses CASTRO (Almgren et al. 2010) with ideal MHD, constrained transport, and the HLLD Riemann solver on a single-level Cartesian grid, with monopole self-gravity and the same zero-temperature equation of state. Equilibria are exported as a vector potential so that $\nabla \cdot \mathbf{B}$ is zero to round-off on the target mesh; the reconstruction is verified before any run, with relative $|\nabla \cdot \mathbf{B}| < 10^{-12}$ and curl errors below 5×10^{-2} .

3. THE FIELD A WHITE DWARF CAN HOLD

3.1. A ceiling independent of central density

We first ask, with rotation switched off, how much confined toroidal field a white dwarf of given central density can carry while satisfying $\beta_{\text{min}} \geq 1$. Figure 1 gives the answer.

The mass that such a field buys is $+0.013$ to $+0.017 M_\odot$, between one and two per cent, and the maximum $E_{\text{tor}}/|W|$ the star holds is 0.0098–0.0101 across a full decade in central density: the ceiling barely moves. For comparison, the configuration that reaches

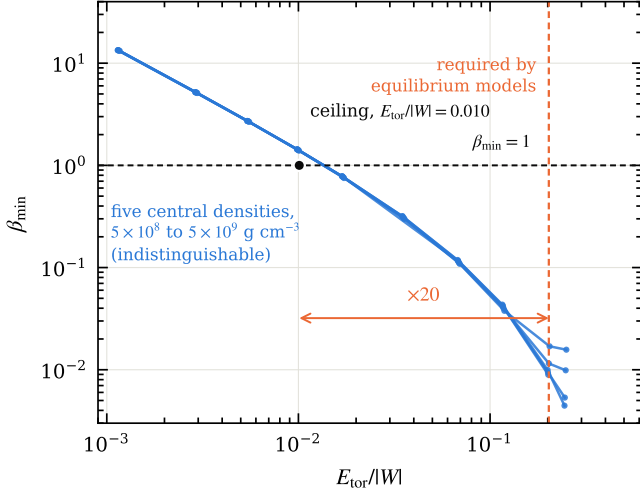


Figure 1. The minimum plasma β over the stellar interior against the energy the confined toroidal field is asked to carry, for five central densities between 5×10^8 and $5 \times 10^9 \text{ g cm}^{-3}$. Each curve is the envelope over the poloidal shape parameters, that is the best β the star achieves at that field strength. The five curves are indistinguishable: the ceiling is a property of the confinement rather than of the star. The star holds its field only above the dashed line, which places the ceiling at $E_{\text{tor}}/|W| = 0.010$, a factor of twenty below the value the super-Chandrasekhar equilibrium models require.

2.007 $M_{\odot}$ at $\rho_c = 10^9 \text{ g cm}^{-3}$ in the standard construction carries $E_{\text{tor}}/|W| = 0.203$ — a factor of twenty above the ceiling.

The reason the ceiling is so low is geometric rather than energetic. A twisted torus requires the toroidal field to live only inside the closed-line region, which is 37–45% of the stellar volume depending on the localization exponent m . Forcing the energy that supports the mass into that fraction of the volume raises the peak field where it lives, and β fails in a shell near $\rho \sim 10^6 \text{ g cm}^{-3}$ before the poloidal component has been introduced at all. We verified this directly: at fixed $E_{\text{tor}}/|W| = 0.203$, $\beta_{\text{min}} = 6 \times 10^{-3}$ and is identical to three digits across three decades of poloidal amplitude. It is the confinement, not the mixed geometry, that breaks β .

3.2. The frontier saturates at the Chandrasekhar mass

Because the ceiling rises slowly with central density while the field-free mass rises toward M_{Ch} , the two effects nearly cancel. Extending the survey to $\rho_c = 1.8 \times 10^{10} \text{ g cm}^{-3}$, just below the neutronization threshold of $1.94 \times 10^{10} \text{ g cm}^{-3}$ for $\mu_e = 2$, the heaviest configuration holding its own confined toroidal field is $1.4387 M_{\odot}$, and the margin to M_{Ch} halves at each density step: +0.0123, +0.0062, +0.0026, +0.0013 $M_{\odot}$.

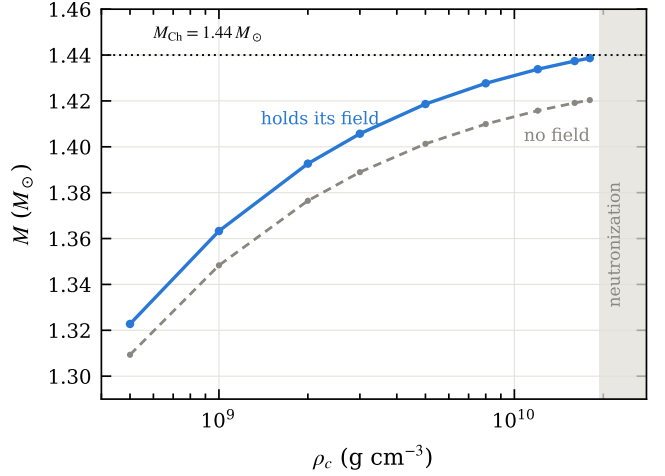


Figure 2. The heaviest configuration that holds its own confined toroidal field (solid) against central density, with the field-free sequence (dashed) for scale. The separation between the curves is what the field buys, between 0.013 and 0.017 $M_{\odot}$. The upper curve approaches the Chandrasekhar mass without reaching it, and the sequence terminates at the neutronization threshold (shaded) rather than at a field limit.

The frontier therefore converges to the Chandrasekhar mass from below and does not cross it. This is a stronger statement than “well below”: a magnetically supported twisted torus saturates *at* the limit that degeneracy alone already provides. Restricted to the comparable-energy band, the heaviest such configuration is $1.4245 M_{\odot}$.

4. WITH ROTATION: THE CONFIGURATION EXISTS

The exclusion of Section 3 is about the *source* of the mass, not about twisted tori. Repeating the survey with differential rotation, 40 of 540 configurations satisfy all four criteria simultaneously.

Mass and comparability are not independent: both rise with rotation, so the $E_{\text{tor}}/E_{\text{pol}}$ criterion binds before the secular bar mode does. The object is therefore a curve rather than a point, and Figure 3 and Table 1 give it. A single quoted maximum would be fragile — the heaviest configuration sits at $E_{\text{tor}}/E_{\text{pol}} \simeq 10$, and whether it falls inside the band depends on where the edge is drawn, which is a convention rather than a physical boundary.

The heaviest configuration satisfying all four criteria is $2.13 M_{\odot}$, $1.48 M_{\text{Ch}}$, with $E_{\text{tor}}/E_{\text{pol}} = 9.62$. Sitting comfortably inside the Braithwaite branch rather than at its edge costs mass but does not cost the result: at $E_{\text{tor}}/E_{\text{pol}} = 4.17$ the star is still $1.65 M_{\odot}$, 14% above the Chandrasekhar limit.

Table 1. Mass against comparability at 257^2 , with $\beta_{\min} = 1$ imposed.

Ω_c/Ω_K	$M (M_\odot)$	$T/ W $	$E_{\text{tor}}/E_{\text{pol}}$	$E_{\text{tor}}/ W $	criteria
1.200	1.6484	0.0606	4.17	0.00594	all
1.350	1.7758	0.0806	5.65	0.00609	all
1.500	1.9623	0.1054	7.50	0.00628	all
1.600	2.1329	0.1246	9.62	0.00645	all
1.675	2.2903	0.1399	10.69	0.00662	fails (iii)

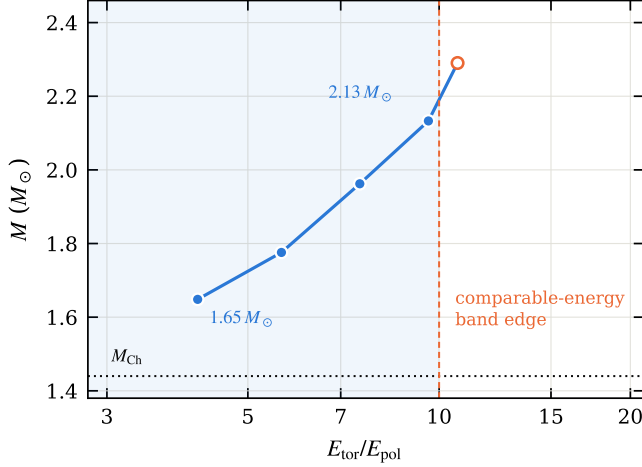


Figure 3. Mass against the ratio of toroidal to poloidal energy along the rotating sequence, with $\beta_{\min} = 1$ imposed at every point. Both quantities rise together with rotation, so criterion (iii) binds before the secular bar mode does; the open symbol is the configuration that falls outside the comparable band. Shading marks the band. The heaviest configuration satisfying every criterion is $2.13 M_\odot$, and a configuration sitting well inside the band rather than at its edge still carries $1.65 M_\odot$, 14% above the Chandrasekhar limit.

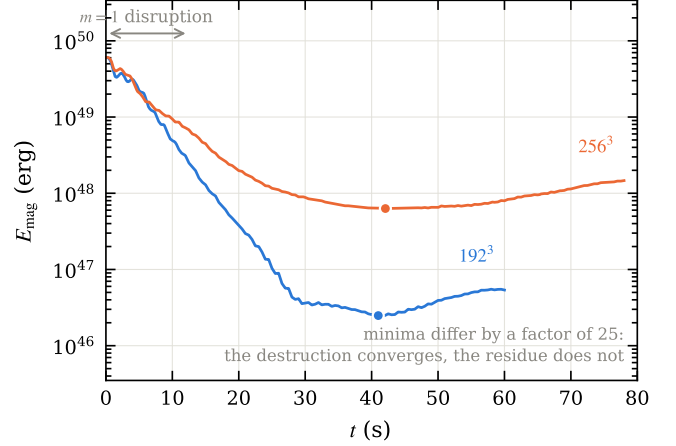


Figure 4. Magnetic energy of the star against time on both meshes. The ordered field is destroyed over the first several seconds and the decay is robust: the two meshes agree on it. Where they do not agree is the level the field settles to, the minima differing by a factor of 25, after which the energy rises again on both. We show both meshes rather than one so that the residue is not mistaken for a measurement; growth rates and residues are reported as resolution-limited throughout (Section 7).

Two features of this result deserve emphasis.

First, the field ceiling does not move. At the mass maximum $E_{\text{tor}}/|W| = 0.0066$, the same one per cent as everywhere else. The additional mass is entirely rotational. Rotation passes the Chandrasekhar limit; the field does not.

Second, rotation does not make β worse, as one might expect from the reduced central pressure of a centrifugally supported star. The measured ceiling rises slightly, from 0.00548 to 0.00628, as Ω_c/Ω_K goes from 0 to 1.5, while the mass nearly doubles.

The upper end of the sequence is set by the secular bar mode, not by the field: $T/|W|$ crosses 0.14 between $\Omega_c/\Omega_K = 1.60$ and 1.70, while the mass-shedding parameter stays near 0.48 throughout.

5. THREE-DIMENSIONAL EVOLUTION

5.1. The literature's configuration is destroyed

Evolving the specified configuration — $2.005 M_\odot$, interior toroidal field of 3.2×10^{13} G beside a 10^9 G exterior dipole, so $B_t/B_p \simeq 4 \times 10^3$ — in three-dimensional ideal MHD, an $m = 1$ instability grows in a few Alfvén times and the ordered field is destroyed (Figure 4): the magnetic energy falls from 6.05×10^{49} to 1.35×10^{49} erg over the first twelve seconds. The star itself and the differential rotation that supports it survive that episode.

This is the behaviour linear theory predicts for a near-pure toroidal field, and it is worth being precise about what it does and does not show. It does not show that toroidal fields cannot exist in white dwarfs; it shows that the particular branch these equilibrium models are built

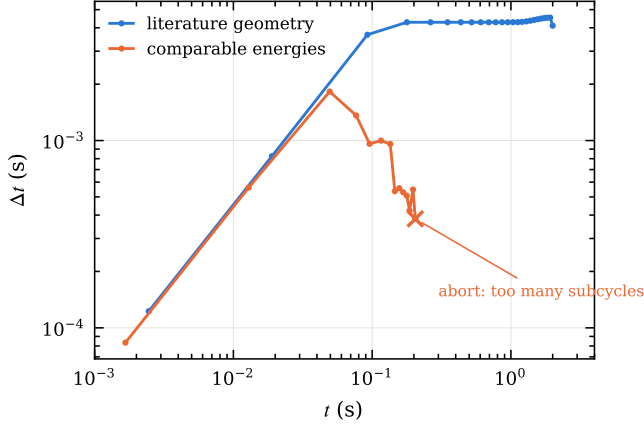


Figure 5. Time step against elapsed physical time for the two configurations, which differ only in poloidal amplitude on an otherwise identical star. Both ramp identically from the initial reduction. The literature geometry then holds a step of a few times 10^{-3} s and reaches the target time; the comparable-energy configuration peaks at $t \simeq 0.05$ s and loses the step in plateaus until the subcycle limit ends the run at $t = 0.205$ s. The pair is asymmetric by construction — only one of the two is ambient-limited — so the figure shows the constraint of Section 5, not a difference in stability.

on is the unstable one. Section 7 lists the caveats that attach to the measured growth rate.

5.2. The stable configuration cannot be evolved in a filled box

We constructed the twisted-torus counterpart on the identical star — same central density, same rotation law, same self-consistent toroidal field, the poloidal amplitude being the only difference — and attempted to evolve both. The control, carrying the literature’s weak exterior dipole, ran cleanly to the target time. The twisted torus aborted at $t = 0.205$ s after a collapse of the time step (Figure 5), at 35 times the cost per unit physical time.

The cause is not numerical fragility but a physical constraint on the setup. Comparable poloidal energy requires an exterior dipole near 7×10^{11} G. An ambient medium of 2×10^4 g cm $^{-3}$, which is already 3% of the stellar mass when it fills the domain, holds at most 3.4×10^{10} G at $\beta = 1$. Raising the ambient to the required 8.7×10^5 g cm $^{-3}$ would put $2.5 M_{\odot}$ of material in the box, more than the star. Stronger current confinement does not help: over the configurations satisfying all four criteria, the Fujisawa localization reduces the exterior dipole by 8% where a factor of 18 is needed.

We therefore report this as a result rather than a difficulty. *The stable branch of magnetized white dwarf equilibria cannot be evolved by a domain-filling code without a vacuum or stratified exterior treatment*, and this is a

plausible reason why the field has equilibrium models and no dynamics.

6. DISCUSSION

The results above are consistent and point one way. A white dwarf can hold only about one per cent of its binding energy in a confined, stably configured magnetic field. That is twenty times less than the super-Chandrasekhar equilibrium programme requires, and it is not enough to lift the mass past degeneracy’s own limit. Where a super-Chandrasekhar magnetized white dwarf is constructible with a stable field geometry, the mass is supplied by differential rotation and the field is a passenger.

This does not refute the existence of over-massive white dwarfs. It relocates their support.

7. LIMITATIONS

We state these here rather than in a closing paragraph because two of them bear directly on how the dynamical results should be read.

The magnetorotational instability is not resolved, and cannot be. With the specified field the fastest-growing MRI wavelength is 4×10^{-3} of a cell. Resolving it is not a question of cost: the requirement that the ambient medium hold the field while the MRI wavelength spans six cells cannot be met by any amplitude within this family. The literature on merger remnants finds the MRI erasing differential rotation when it is resolved (Pakmor et al. 2024). Our differential rotation survives, and that outcome is not distinguishable from our simulation being unable to destroy it.

The $m = 1$ mode grows on grid noise. The intended initial poloidal seed is $E_{\text{pol}}/E_{\text{tor}} = 4.6 \times 10^{-8}$; the value realized on the mesh is 1.9×10^{-6} , forty times larger, and the difference is discretization error. The Tayler instability is ideal, so the seed sets the onset time rather than whether the mode exists, and the growth rate, the $m = 1$ character and the angular-momentum transport are robust to it. The measured growth rate is not: it rises by 47% from 192^3 to 256^3 , and the late-time field residue differs by a factor of 25 between the same meshes. Growth rates and residues are reported as resolution-limited.

Part of the evolution lies above the critical field. In the first twelve seconds, 49% of sampled instants have a peak field exceeding the Landau critical value, where a field-independent degenerate equation of state is least defensible. The rotation results, measured from 12 s onward, are free of this.

$\beta_{\text{min}} \geq 1$ is a consistency criterion. It states that the equilibrium does not contradict the equation of state

used to build it. It is not a stability proof, and only three-dimensional evolution can supply one.

The strongly localized corner is not converged.

For $m \leq -1.5$ the survey quantities shift by up to 2.4% in $\beta_{\min}$ and 13% in $E_{\text{tor}}/E_{\text{pol}}$ between 129^2 and 257^2 , and the shift does not diminish with refinement. No result reported here draws on that corner: the configurations satisfying all four criteria use $m = 0$ and $m = -1$, where the same quantities move by 0.03% and 1% respectively.

8. CONCLUSIONS

1. A white dwarf holds at most $E_{\text{tor}}/|W| \simeq 0.0101$ in a confined toroidal field, nearly independent of central density. The super-Chandrasekhar equilibrium programme requires twenty times more.
2. Such a field is worth about $+0.017 M_{\odot}$. The mass frontier it permits converges to M_{Ch} from below and does not cross it, ending at neutronization.
3. With differential rotation, twisted-torus configurations above the Chandrasekhar mass do exist, reaching $2.13 M_{\odot}$ at the edge of the comparable band and $1.65 M_{\odot}$ well inside it, but their support

is rotational and the field still carries about one per cent of the virial.

4. The configuration the literature specifies is destroyed by an $m = 1$ instability in a few Alfvén times, as linear theory predicts for a near-pure toroidal field.
5. The stable alternative cannot be evolved in a domain-filling code: comparable poloidal energy implies an exterior dipole that no affordable ambient medium confines.

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